Lesson 11: Wave After Wave

GDevelop

VIDEO GAME CREATION

This week: clearing a wave of enemies no longer means the game is over - it means the NEXT wave begins. Clear all 3, and that's when you really win.

Open your own [YourName]Game project from Lesson 10 - don't start a new one.

Tip: no new assets this week - multiple waves reuse your exact existing enemy spawn logic. This completes the Level 2 Complete milestone!

Setup

1. Open your [YourName]Game project from Lesson 10.
2. Add two new scene variables: Wave (default 1) and TotalWaves (default 3).
3. Add a new Text object, WaveText, visible, positioned near ScoreText/LivesText.

Level 1 - Minimal (MVP)

Clearing a wave's enemies spawns a fresh wave, up to 3 total. Clearing wave 3 triggers the real win. The current wave number is shown on screen.

4. In the Setup event group (from Lesson 10), add Wave = 1 alongside the other resets.
5. In the HUD always-event, add an action changing WaveText's content to show Wave and TotalWaves.
6. Split your Combat group's win-check into two separate events (code below): one for "this was the last wave" (the real win), one for "more waves remain" (advance and respawn).
7. Click Preview and play through - clearing wave 1 or 2 should spawn a fresh wave instead of ending the game; clearing wave 3 should show WinText.

```
(Combat event group, revised)
Condition: Bullet is in collision with Enemy
Action: Play a sound (sfx_hit.wav)
Action: EnemiesRemaining = EnemiesRemaining - 1
Action: Score = Score + 10
Action: Delete Bullet
Action: Delete Enemy

(new event - full win)
Condition: EnemiesRemaining <= 0
Condition: Wave >= TotalWaves
Action: GameActive = 0
Action: Show WinText
```

```
(new event - advance wave)
Condition: EnemiesRemaining <= 0
Condition: Wave < TotalWaves
Action: Wave = Wave + 1
Action: EnemiesRemaining = 5
Action: Repeat 5 times -> Create Enemy...
```

Level 2 - Complete (Alpha/Beta)

Wave transitions are seamless and correct every time - no double-spawns, no skipped waves, exactly 3 waves before the real win, and restarting always returns to wave 1.

- Play a full 3-wave game: confirm waves 1 and 2 each spawn a fresh set of enemies with no win message, and only wave 3 triggers WinText.
- Let Lives run out partway through wave 2 and confirm LoseText still works correctly, not a wave transition.
- Press R after a win or loss and confirm Wave resets to 1, along with Score and Lives.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: briefly show a "Wave 2!" style message using a separate Text object with a short Timer-based hide action, distinct from WinText/LoseText.
- Structure: notice the "Repeat 5 times -> Create Enemy" action now appears in two places (Setup, and the advance-wave event) - for extra practice, try extracting it into a Function (Project > Functions) callable from both, if you're comfortable exploring that feature.

If Something's Wrong

- Wave 1 immediately shows WinText: confirm the full-win event has BOTH conditions (EnemiesRemaining <= 0 AND Wave >= TotalWaves), not just the first.
- New waves never spawn: confirm the advance-wave event's Wave < TotalWaves condition is using the correct comparison direction.
- Both the full-win and advance-wave events fire on the same frame: confirm their Wave conditions are opposites (>= vs. <) so only one can ever be true at a time.

Want More?

Student Resource List > GDevelop section, or gdevelop.io/tutorials